Code No: 118ET **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year II Semester Examinations, July - 2019
STORAGE AREA NETWORKS
(Common to CSE, IT)

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART - A

(25 Marks)

1.a) Write a short note on data management. [2]
b) What are the key metrics to monitor for different components in storage infrastructure? [3]
c) What are the components of FC-SAN? [2]
d) Write a short note on RAID 1 and RAID 2. [3]
e) Write a short note on planned outages. [2]
f) Explain briefly about Disaster Recovery. [3]
g) Explain the Key management tasks in a data center. [2]
h) Write a short note on Backup Recovery. [3]
i) What is meant by Storage security domain? [2]
j) Write a short note on virtualization technologies. [3]

PART - B

(50 Marks)

2. What are the elements of a data center infrastructure? Describe the role of each element in supporting business activities. [10]

OR

3. Briefly explain about the challenges in data storage and data management. [10]

4.a) Explain working of an intelligent storage system with a neat architecture.
b) Explain about RAID Array Components. [5+5]

OR

5.a) Briefly explain about the components of NAS.
b) Explain the benefits of the different networked storage options for different application environments. [5+5]

6.a) Briefly explain about unplanned outages.
b) Discuss various solutions to mitigate single points of failure in a storage infrastructure. [5+5]

OR

7. Explain about the following
a) RTO b) RPO. [5+5]

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8.a) What is the role of replication technologies in ensuring information availability?
b) Explain the Architecture of backup / recovery. [5+5]

OR

9. Explain the different backup/recovery topologies. [10]

10.a) What are the key management tasks in virtualization?
b) Describe Industry standards for data center monitoring and management. [5+5]

OR

11. Explain about following
a) Information Security b) Journaling. [10]

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